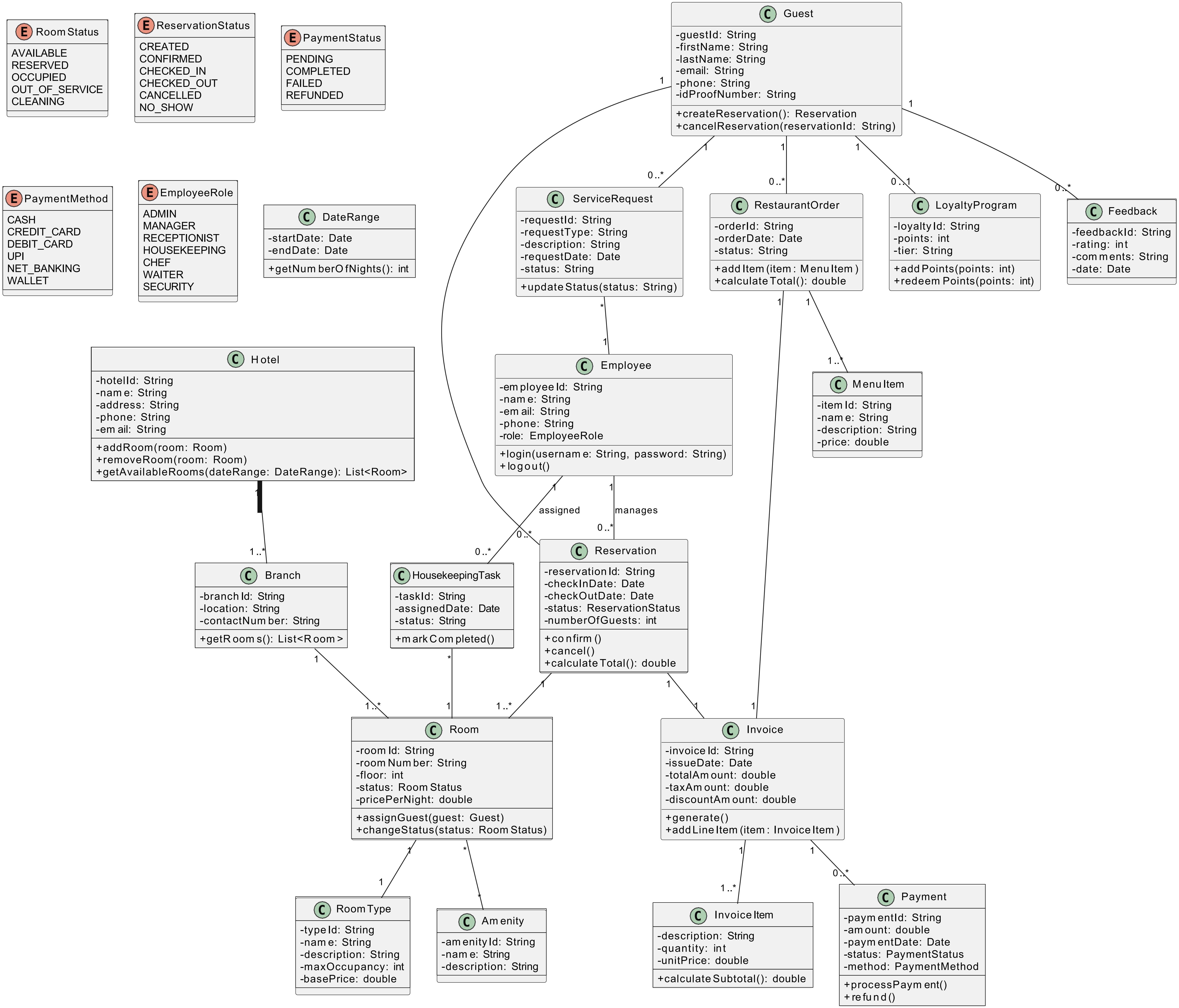
Hotel Management System - Detailed Class Diagram



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HOTEL MANAGEMENT SYSTEM
Detailed Class Diagram Description

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1. Introduction

The Hotel Management System is designed to manage hotel operations efficiently, including room booking, guest management, billing, employee operations, restaurant services, housekeeping, loyalty programs, and feedback management.
The system ensures smooth coordination between guests, employees, rooms, reservations, payments, and services.

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2. Enumerations

RoomStatus

- AVAILABLE
- RESERVED
- OCCUPIED
- OUT_OF_SERVICE
- CLEANING

ReservationStatus

- CREATED
- CONFIRMED
- CHECKED_IN
- CHECKED_OUT
- CANCELLED
- NO_SHOW

PaymentStatus

- PENDING
- COMPLETED
- FAILED
- REFUNDED

PaymentMethod

- CASH
- CREDIT_CARD
- DEBIT_CARD
- UPI
- NET_BANKING
- WALLET

EmployeeRole

- ADMIN
- MANAGER
- RECEPTIONIST
- HOUSEKEEPING
- CHEF
- WAITER
- SECURITY

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3. Core Classes and Attributes

3.1 Hotel

Attributes

- hotelId : String
- name : String
- address : String
- phone : String
- email : String

Methods

- addRoom(room : Room)
- removeRoom(room : Room)
- getAvailableRooms(dateRange : DateRange) : List<Room>

Relationships

- A Hotel contains multiple Branches (1..*)
- A Hotel manages multiple Rooms

3.2 Branch

Attributes

- branchId : String
- location : String
- contactNumber : String

Methods

- getRooms() : List<Room>

Relationships

- A Branch belongs to one Hotel
- A Branch contains multiple Rooms

3.3 Room

Attributes

- roomId : String
- roomNumber : String
- floor : int
- status : RoomStatus
- pricePerNight : double

Methods

- assignGuest(guest : Guest)
- changeStatus(status : RoomStatus)

Relationships

- A Room belongs to one Branch
- A Room has one RoomType
- A Room can have multiple Amenities
- A Room can be associated with multiple Reservations over time

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3.4 RoomType

Attributes

- typeId : String
- name : String
- description : String
- maxOccupancy : int
- basePrice : double

Relationships

- One RoomType can be associated with multiple Rooms

3.5 Amenity

Attributes

- amenityId : String
- name : String
- description : String

Relationships

- A Room can have multiple Amenities

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4. Guest Management

4.1 Guest

Attributes

- guestId : String
- firstName : String
- lastName : String
- email : String
- phone : String
- idProofNumber : String

Methods

- createReservation() : Reservation
- cancelReservation(reservationId : String)

Relationships

- A Guest can have multiple Reservations
- A Guest can submit Feedback
- A Guest can enroll in one LoyaltyProgram
- A Guest can create multiple ServiceRequests
- A Guest can place multiple RestaurantOrders

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5. Reservation & Billing

5.1 Reservation

Attributes

- reservationId : String
- checkInDate : Date
- checkOutDate : Date
- status : ReservationStatus
- numberOfGuests : int

Methods

- confirm()
- cancel()
- calculateTotal() : double

Relationships

- A Reservation is linked to one Guest
- A Reservation is linked to one Room
- A Reservation generates one Invoice
- Managed by an Employee

5.2 DateRange

Attributes

- startDate : Date
- endDate : Date

Methods

- getNumberOfNights() : int

5.3 Invoice

Attributes

- invoiceId : String
- issueDate : Date
- totalAmount : double
- taxAmount : double
- discountAmount : double

Methods

- generate()
- addLineItem(item : InvoiceItem)

Relationships

- One Invoice belongs to one Reservation
- One Invoice contains multiple InvoiceItems
- One Invoice has one or more Payments

5.4 InvoiceItem

Attributes

- description : String
- quantity : int
- unitPrice : double

Methods

- calculateSubtotal() : double

5.5 Payment

Attributes

- paymentId : String
- amount : double
- paymentDate : Date
- status : PaymentStatus

- method : PaymentMethod

Methods

- processPayment()
- refund()

Relationships

- Payment is linked to one Invoice

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6. Employee Management

6.1 Employee

Attributes

- employeeId : String
- name : String
- email : String
- phone : String
- role : EmployeeRole

Methods

- login(username : String, password : String)
- logout()

Relationships

- Employee manages Reservations
- Employee handles ServiceRequests
- Employee assigned HousekeepingTasks

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7. Additional Services

7.1 HousekeepingTask

Attributes

- taskId : String
- assignedDate : Date
- status : String

Methods

- markCompleted()

Relationships

- Assigned to one Employee
- Related to one Room

7.2 ServiceRequest

Attributes

- requestId : String
- requestType : String
- description : String
- requestDate : Date
- status : String

Methods

- updateStatus(status : String)

Relationships

- Requested by Guest
- Managed by Employee

7.3 RestaurantOrder

Attributes

- orderId : String
- orderDate : Date
- status : String

Methods

- addItem(item : MenuItem)
- calculateTotal() : double

Relationships

- Placed by Guest
- Contains multiple MenuItems

7.4 MenuItem

Attributes

- itemId : String
- name : String
- description : String
- price : double

7.5 LoyaltyProgram

Attributes

- loyaltyId : String

- points : int
- tier : String

Methods

- addPoints(points : int)
- redeemPoints(points : int)

Relationships

- One LoyaltyProgram per Guest

7.6 Feedback

Attributes

- feedbackId : String
- rating : int
- comments : String
- date : Date

Relationships

- Submitted by Guest
- Linked to Reservation

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8. Relationships Summary

- Hotel → Branch (1 to many)
- Branch → Room (1 to many)
- Room → Reservation (1 to many over time)
- Guest → Reservation (1 to many)
- Reservation → Invoice (1 to 1)
- Invoice → Payment (1 to many)
- Reservation → Room (many to 1)
- Employee → Reservation (1 to many)
- Room → Amenity (many to many)

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9. Conclusion

The Hotel Management System class diagram demonstrates:

- Proper use of inheritance and structured modelling
- Use of enumerations for system states
- Strong cohesion between booking, billing, and service modules
- Clear multiplicity relationships
- Logical separation of operational responsibilities